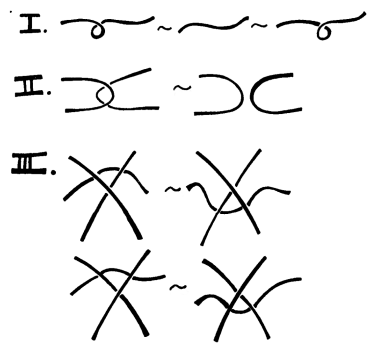
Knot and Link Invariants

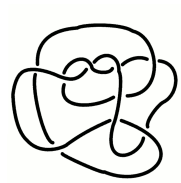
1. Introduction

A **knot** is a circle "embedded" in three-dimensional space. A **link** is a collection of several knots. We say two knots (or two links) are equivalent, denoted ~, if we can smoothly deform one into the other.

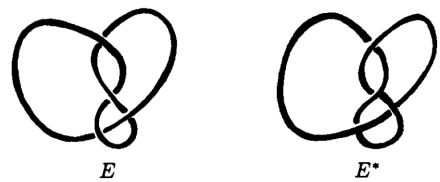


We consider a **diagram** of a knot by projecting it onto a plane and indicating under/over at each crossing. Reidemeister proved that two knots are equivalent if and only if their corresponding diagrams can be transformed into each other using a sequence of the three **Riedemeister moves** shown above.

Problem 1.1. The standard circle ◯ is called the **unknot**. Let K be the knot below. Is it true ◯ ~ K? If so, draw a sequence of Reidemeister moves (I,II,III) transforming K to the unknot.



Problem 1.2. Let E be the figure-eight knot and consider its mirror image E^* (reverse all the crossings). Draw a sequence of Riedemeister moves to show $E \sim E^*$.



Problem 1.3. Let K be any knot and K^* its mirror image. Is it always true that $K \sim K^*$? Justify your answer.

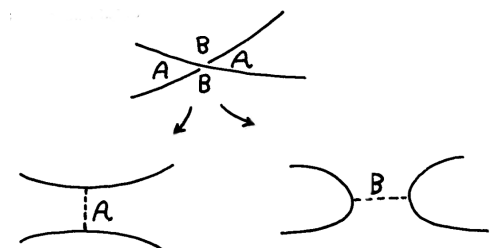
Problem 1.4 (Challenge)**.** Give a sequence of Reidemeister moves to show this knot is equivalent to the unknot:

We can give any link an **orientation** by assigning a direction to each link component. We draw this orientation with an arrow. We also record whether each crossing is "positive" or "negative:"

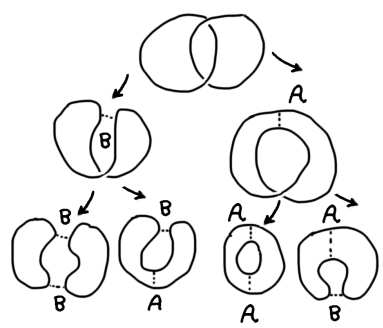
Problem 1.5. The **writhe** of an oriented link diagram is the number of positive crossings minus the number of negative crossings. Show that the writhe of a link diagram is invariant under Reidemeister moves II and III, but not I.

2. Kauffman Bracket

Goal: construct a polynomial for each knot diagram that is invariant under Riedemeister moves. We start by cutting our knot (or link) K into simple pieces. At each crossing in our knot diagram, we make two new **descendant** links by performing a type-A or type-B splicing:



We keep track of each splicing and its type by labeling it A or B. Continue this way until we reach the **primitive descendants**, the descendants with no crossings. For example, we calculate the four primitive descendants of the "Hopf link":



Define the **Kauffman bracket** $\langle K \rangle$ as follows:

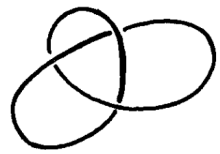
(1) If P is a primitive descendant of K, define the bracket $\langle K|P \rangle$ as the product of all labels of P[1]. For example, the leftmost primitive descendant of the Hopf link has bracket $\langle K|P \rangle = B^2$.
(2) Let $\|P\|$ be the number of components (disjoint knots) in P, minus 1.
(3) Define
$$\langle K \rangle = \sum_P \langle K|P \rangle d^{\|P\|}$$
where the sum is over all primitive descendants P and d is another variable.

Problem 2.1. Check using the definition that the Kauffman bracket of the Hopf link is $B^2 d^1 + 2ABd^0 + A^2 d^1 = (A^2 + B^2)d + 2AB$

[1]We treat A and B as commuting variables

Problem 2.2. Let K be a knot. What is $\langle \bigcirc K \rangle$, where $\bigcirc K$ is the knot formed by adding a disjoint unknot to K?

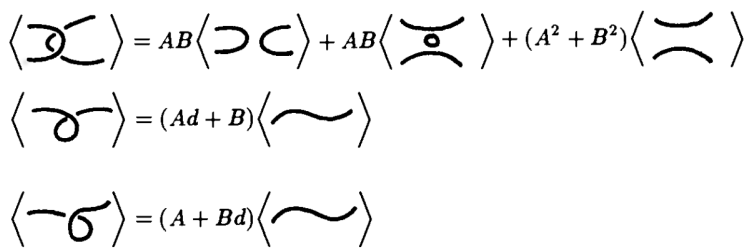
Problem 2.3. Let T be the trefoil knot shown below. Compute $\langle T \rangle$.



Problem 2.4. Use the definition to prove that

Here, the notation means the diagram in the bracket is part of a larger diagram, and we only change the part shown in the bracket.

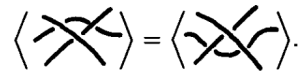
Problem 2.5 (Riedemeister II Invariance)**.** (a) Prove the following identities:



(b) Declare $A = B^{-1}$ and $d = -A^2 - A^{-2}$. With these equalities in place, show that the bracket becomes invariant under Reidemeister II moves.

From now on, we assume $A = B^{-1}$ and $d = -A^2 - A^{-2}$.

Problem 2.6 (Reidemeister III Invariance)**.** Show that the bracket is invariant under Reidemeister III moves, i.e.



To make the bracket invariant under Reidemeister I, we need to modify it further. Let K be an *oriented* link and define the **normalized bracket** of K as

$$\mathcal{L}_K(A) = (-A^3)^{-w(K)}\langle K\rangle$$

where $w(K)$ is the writhe of K (see Problem 1.5).

Problem 2.7. Prove that the normalized bracket is invariant under Reidemeister I moves. Conclude that the normalized bracket is an invariant under all Reidemeister moves and hence determines an oriented knot invariant!

Problem 2.8.
(a) Determine the relationship between $\mathcal{L}_K(A)$ and $\mathcal{L}_{K^*}(A)$.
(b) Compute $\langle T^*\rangle$ where T^* is the mirror image of the trefoil.
(c) Compare with $\langle T\rangle$ to conclude that $T \not\sim T^*$.

Problem 2.9. Consider the collection of knots K_n:

(a) Find a recursive formula for $\langle K_n\rangle$.
(b) Show that K_n is not equivalent to K_n^* for any $n > 1$.

3. Jones Polynomial

The Jones polynomial is another (oriented) knot invariant $V_K(t)$ discovered by Vaughan Jones in quite a different context. Its properties are:

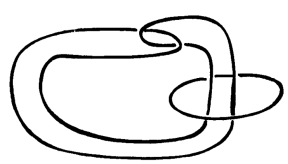
(a) $V_{\circlearrowright}(t) = 1$
(b) If $K \sim K'$ then $V_K(t) = V_{K'}(t)$.
(c) $t^{-1}V_{L_+}(t) - tV_{L_-}(t) = \left(t^{1/2} - t^{-1/2}\right)V_{L_0}(t)$ where L_+, L_-, L_0 are as below:

L_+ L_- L_0

The existence of V_K is not obvious; the following problem relates it to our earlier construction.

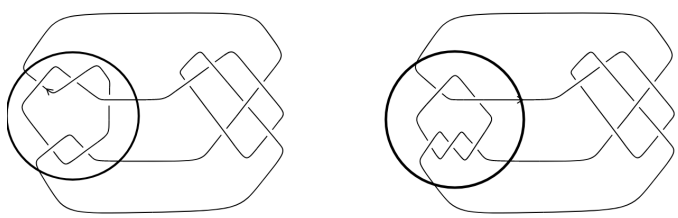
Problem 3.1. Show that $V_K(t) = \mathcal{L}_K(t^{-1/4})$ satisfies (a)-(c)

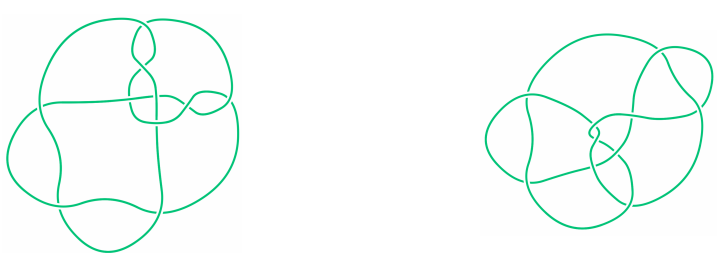
Problem 3.2. Let W be the Whitehead link shown below. Prove that $W \not\sim \bigcirc\bigcirc$ by choosing an orientation and computing the Jones polynomial.



Problem 3.3. Prove using either the Jones polynomial or the Kauffman bracket that the following not is not equivalent to the unknot.

Problem 3.4 (Challenge)**.** Unfortunately, Jones' polynomial cannot always detect nonequivalent knots. It turns out knots (a) and (b) are not equivalent. The transformation shown in the circled region is known as a **Conway mutation**. Prove that the Conway mutation does not change the Jones polynomial of knots (a) and (b).



(a) Kinoshita-Terasaka knot (b) Conway knot

Problem 3.5 (Challenge)**.** Choose an orientation of the following two knots and compute their Jones polynomials. Are they equivalent knots?
(For the answer see knotplot.com/perko/)